Weekly Progress Report - Week 1

Improve performance of existing online trackers for small object detection [UAV Videos]

Group: The Convolutioners
{Aagam Sheth, Mahima Parekh, Khevn Thanki, Riddhi Bhargava}

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Outline of Performed Tasks:

- Reviewed reference materials provided by TA including CVPR 2025 paper, MOT metrics documentation, and GitHub repositories for Kalman Filter implementation, and BoxMOT framework
- Successfully downloaded and explored the AU Drone Dataset
- Studied challenges specific to small object detection in UAV videos such as low pixel resolution, scale variation, background clutter, motion blur, and occlusions

Project Overview

Problem Statement: Improve existing online trackers for small object detection in UAV videos. This study evaluates Computer Vision methods for small object tracking in UAV videos.

Small object detection in UAV imagery presents unique challenges because objects often occupy a very small fraction of the frame (sometimes less than 1% of total pixels). Due to downsampling in convolutional neural networks, fine-grained spatial features may be lost, reducing localization accuracy. Additionally, cluttered backgrounds and dense multi-object scenes increase false positives and identity switches during tracking.

The objective is to design a robust pipeline that enhances detection quality for small objects and improves tracking consistency using online trackers evaluated through standard MOT metrics.

Reference Materials Studied

1. Primary research paper from CVPR 2025:
   `https://cvpr.thecvf.com/virtual/2025/poster/35174`

   Studied advancements in small object detection and multi-object tracking, focusing on feature representation, motion modeling, and robustness in aerial scenarios.

2. MOT tracking metrics documentation:
   `https://miguel-mendez-ai.com/2024/08/25/mot-tracking-metrics`

   Reviewed evaluation metrics including:

   - MOTA (Multiple Object Tracking Accuracy)
   - MOTP (Multiple Object Tracking Precision)
   - IDF1 (Identity F1 Score)

- False Positives (FP) and False Negatives (FN)

3. GitHub repositories for implementation reference:
   - Multiple Object Tracking using Kalman Filter:
     `https://github.com/NickNair/Multiple-Object-Tracking-using-Kalman-Filter`
     Understood motion prediction using state-space models and covariance updates.
   - BoxMOT tracker framework:
     `https://github.com/mikel-brostrom/boxmot`
     Explored integration of detection models (e.g., YOLO) with tracking algorithms such as DeepSORT and ByteTrack.

Outcomes:

- Understood classical tracking methods (SORT, Mean-Shift, CamShift) and Kalman Filtering
- Understood AU Drone Dataset structure including annotation format and video sequences
- Identified key challenges in small object detection affecting tracker performance
- Understood relationship between detection accuracy and tracking metrics

Tentative List of Tasks for Next Week:

1. Set up development environment (Python, OpenCV, PyTorch/TensorFlow, YOLO v8) and complete study of reference papers
2. Understand MOT evaluation metrics (MOTA, MOTP, IDF1)